Dipon Deb Dipu

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PROFILE

Embedded systems master's student on a full Uppsala University Global Scholarship. I write C and C++ close to the hardware, verify it formally with Frama-C, and work with real-time systems, wireless embedded protocols and computer architecture; this term I am adding FPGA design and model-based control. Co-author of two IEEE conference papers on deep learning for medical imaging. Looking for a master's thesis position in 2027.

EDUCATION

M.Sc. in Embedded Systems, Uppsala University, Sweden — Sep 2025 – Jun 2027 (expected)

- Uppsala University Global Scholarship (full).
- Courses include Real Time Systems (Ada, UPPAAL), Advanced Computer Architecture, Programming Embedded Systems (Zephyr, Frama-C/ACSL), Wireless Communication and Networked Embedded Systems, Microcontroller Programming, Introduction to Parallel Programming (OpenMP, MPI).
- In progress, autumn 2026: Accelerating Systems with Programmable Logic (Verilog, AMD Vivado), Model-Based Design of Embedded Software (MATLAB, Simulink, Stateflow, PID and LQR), Machine Learning for the Internet of Things.

B.Sc. in Computer Science and Engineering, Ahsanullah University of Science and Technology, Dhaka — Jan 2020 – Apr 2024

- CGPA 3.752 / 4.00, ranked 6th of 93 graduates. Dean's List of Honor.
- A+ in Microprocessors, Digital Logic Design, Digital Electronics, Operating Systems, Algorithms, Database, and Project and Thesis I and II.

EXPERIENCE

Backend Developer and System Administrator, Arthur Private Company Ltd., Dhaka — Feb 2023 – Jun 2025

Part-time alongside the B.Sc. (Feb 2023 – Mar 2024), then full-time. A trading and vendor company that bids on tenders and manages inventory.

- Built the company website (HTML5, JavaScript).
- Built and maintained the backend inventory system on an Oracle database.
- Administered the company's systems and wrote Python automation scripts.

PROJECTS

Wireless Backscatter and Network-Wide Consensus Protocol · Uppsala team project — 2026

- Planned and coded software-defined forward error correction and adaptive bitrate switching for a low-power backscatter protocol; physical-layer packet measurements and power tuning. C/C++, nRF52840, CC1352P7, Raspberry Pi.

Formally verified embedded C · Uppsala coursework — 2026

- Sorting, searching and more written in C/C++ with ACSL specifications and proved with Frama-C, including the final embedded project.

Vision-controlled 19-DoF robotic hand · personal — 2026

- Real-time pipeline mapping 21 MediaPipe hand landmarks to the 19 actuators of a TIAGo Hey5 hand in Webots, with vector kinematics to avoid mesh clipping and a low-latency UDP video stream. Python, OpenCV.

gem5: CXL-attached NPU and LLaMA offload · self-directed learning — 2026

- Learned the gem5 full-system simulator by modelling a CXL-attached NPU and offloading LLM work to it. C++, Python.

Digital hardware design and simulation · AUST coursework — 2021 – 2023

- ALUs, a Booth multiplier and SAP computers in Proteus 8; 8086 assembly programming.

PUBLICATIONS

- S. Barua, **D. D. Dipu**, T. Broti, "A Hybrid Classical-Quantum Deep Learning Framework for MRI-Guided Alzheimer's Disease Classification, Comparative Analysis & Explainable AI for Medical Interpretation," IEEE INDISCON 2025, Rourkela, India. DOI 10.1109/INDISCON66021.2025.11252366. Implemented the hybrid CNN and 8-qubit quantum model in PyTorch and PennyLane (96.26% accuracy) and the Grad-CAM explanations.
- S. Barua, **D. D. Dipu**, T. Broti, "NeuroMindscape: Deep Learning Unraveling Alzheimer's Stages Through MRI-Based Classification," IEEE PEEIACON 2024, Rajshahi, Bangladesh. DOI 10.1109/PEEIACON63629.2024.10800075. Designed the 7-layer CNN (97.05% accuracy with augmentation).

SKILLS

Embedded	C, C++, Embedded C, Ada; timers, interrupts, serial I/O; Zephyr, TinyOS, Contiki; scheduling, WCET, state machines
Verification	Frama-C with ACSL, UPPAAL, Simulink Design Verifier
Hardware	nRF52840, CC1352P7, Raspberry Pi; Verilog and AMD Vivado (in progress); Proteus, 8086 assembly
Architecture	Caches, coherence, pipelines, out-of-order and VLIW, GPUs; gem5; OpenMP, MPI
Wireless / IoT	Backscatter, FEC, adaptive bitrate, medium access, Cooja, UDP/TCP
ML and vision	PyTorch, Keras, PennyLane, CNNs, Grad-CAM, OpenCV, MediaPipe
Tools and web	Git, Linux, Jupyter, Jira, MATLAB/Simulink; Python, SQL, Oracle Database, HTML5, JavaScript

LEADERSHIP

- AUST CSE Society: General Secretary, then Vice President (2022 – 2024). Ran departmental events and showcases, including hardware logistics.
- AUST Research and Publication Club: founding member and Vice President; organised the 2023 research symposium for students and faculty.

LANGUAGES

Bengali (native) · English (fluent, professional) · Swedish (learning)